Def. A function $f: G \to \mathbb{C}$ defined on an open set $G \subseteq \mathbb{C}$ is called Meromorphic if for some finite number points $z_1, \cdots, z_n \in G$,

f is Holomorphic on $G \setminus \{z_1, \cdots, z_n\}$ and f has pole at $z \in \{z_1, \cdots, z_n\}$.

Thm. Every Meromorphic function on Extended Complex plane is Rational function.

Proof Let f be a Meromorphic function on Extended Complex plane.
That is, f is Meromorphic on $\mathbb{C}$ and the function $z \mapsto f(1/z)$ has a pole or removable singular at $z=0$.
Let $z_1, \cdots, z_n$ be poles in $\mathbb{C}$ of $f(z)$. Then, for each $1 \le k \le n$,
there exists some neighbor B_k and f_k, g_k such that

$$f(z) = f_k(z) + g_k(z),$$

where f_k is principal part of f at z_k and g_k is analytic at z_k.
And, similarly, for some neibor of $z=0$, there are $\tilde{f}_\infty, \tilde{g}_\infty$

$$f\left(\frac{1}{z}\right) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$$

where ...
Finally, let $f_\infty(z) = \tilde{f}_\infty\left(\frac{1}{z}\right)$. Define $H: \mathbb{C} \to \mathbb{C}$ as

$$H(z) = f(z) - f_\infty(z) - \sum_{k=1}^{n} f_k(z)$$

Then, for each $1 \le k \le n$, z_k is removable singularity of H,
and $H(1/z)$ has removable singularity at $z=0$, therefore it is bounded in some neighbor of $z=0$, thus H is bounded and entire, so that by the Liouville's theorem, H is constant. Now,

$$f(z) = H + f_\infty(z) + \sum_{k=1}^{n} f_k(z). \qquad \square$$